Module 5 (b) - Thyristors

Thyristors: The four layer Diode, SCR, SCR Phase control, Bidirectional Thyristors, IGBTs, Other Thyristors

Thyristors

A Thyristor is a semiconductor device that uses internal feedback to produce switching action.The most important Thyristors are Silicon Controlled Rectifier (SCR) and the Triac.

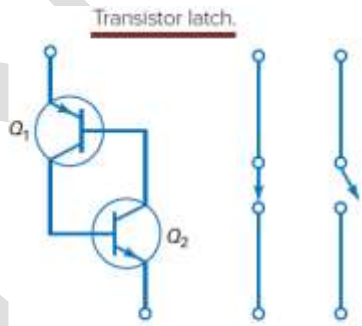
Applications

Thyristors are used for (i) overvoltage protection (ii) motor controls, (iii) heaters (iv) lighting systems, and other heavy current loads.

The Four-Layer Diode (Shockley Diode)

The upper transistor Q1 is a pnp device and the lower transistor Q2 is an npn device. The collector of Q1 drives the base of Q2. Similarly, the collector of Q2 drives the base of Q1.

Principle



The Circuit uses positive feedback. Any change in the base current of Q2 is amplified and fed back through Q1 to magnify the original change. This positive feedback continues changing the base current of Q2 until both transistors go into either saturation or cutoff.

Working

(i) As closed Switch - If the base current of Q2 increases, the collector current of Q2 increases. This increases the base current of Q1 and the collector current of Q1. More collector current in Q1 will further increase the base current of Q2. This amplify-and-feedback action continues until both transistors are driven into saturation. In this case, the overall circuit acts like a closed switch.

(ii) As Open Switch - If something causes the base current of Q2 to decrease, the collector current of Q2 decreases, the base current of Q1 decreases, the collector current of Q1 decreases, and the base current of Q2 decreases further. This action continues until both transistors are driven into cutoff. Then, the circuit acts like an open switch.

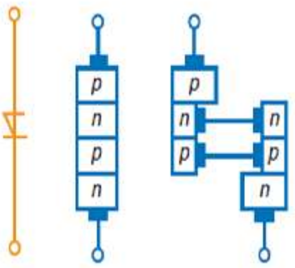
Conclusion

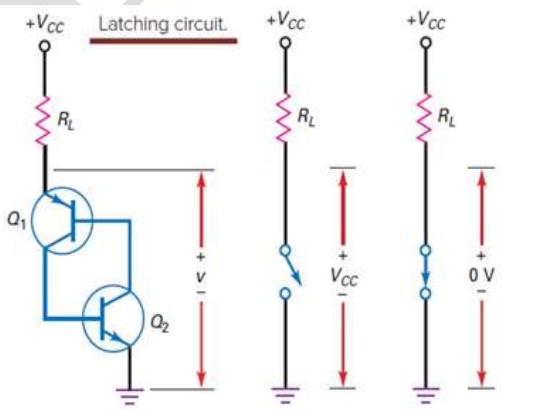
The circuit is stable in either of two states: open or closed. It will remain in either state indefinitely until acted on by an outside force. Because the circuit can remain in either state indefinitely, it is called a latch.

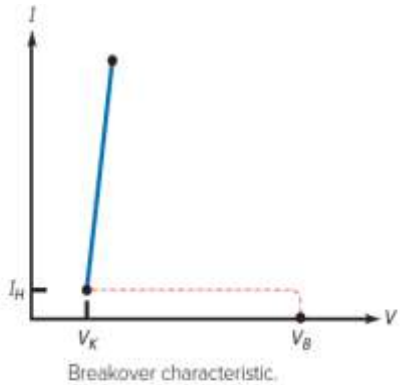
Closing the Latch (Applying Break-over Voltage)

Figure shows a four layered diode connected to a load resistor with a supply voltage of $V_{CC.}$ Assume that the latch is open, because there is no current through the load resistor, the voltage across the latch equals the supply voltage. The only way to close the latch is by using a large enough supply voltage V_{CC} to break down the Q1 collector diode called as **Break-over.**

Since the collector current of Q1 increases the base current of Q2, the positive feedback will start.







This drives both transistors into saturation. When saturated, both transistors ideally look like short-circuits, and the latch is closed, the latch has zero voltage across it when it is closed.

Opening a Latch

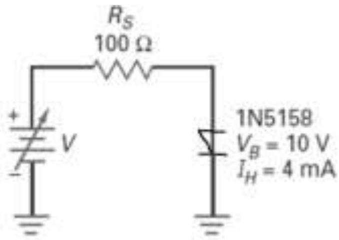
By reducing the V_{CC} supply to zero. This forces the transistors to switch from saturation to cutoff, this type of opening is called **low-current drop-out** because it depends on reducing the latch current to a value low enough to bring the transistors out of saturation.

> The only way to close a four-layer diode is by breakover and open it is by low-current drop-out (reducing the current to less than the holding current).

Holding Current: Thyristor can be turned off or can return to forward blocking state if and only if current through Thyristor fall below certain value of current called as Holding current.

Latching Current: It is the minimum value of anode current which is necessary to make SCR in conduction mode even though gate pulse are removed.

Problem (a) The diode has a break-over voltage of 10 V. If the input voltage of is increased to 15V, what is the diode current? (b) If the diode has a holding current of 4 mA. The input voltage is increased to 15 V to latch the diode and then decreased to open the diode. What is the input voltage that opens the diode?



Solution

(a) Given V_{in} = 15V; V_{BO} = 10 and $V_{in} > V_{BO}$

The diode breaks over. Under ideal condition $I = (V_{in} / R) = (15/100) = 150$ mA

Using the approximation as V_k= 0.7V $I = (V_{in} – V_k)/ R = (15 – 0.7)/100 = 143$ mA

(b) The holding current, given as 4 mA.

At this small current, the diode voltage is approximately equal to the knee voltage, 0.7 V. Since 4 mA flows through 100 Ω, the input voltage is:

- Vin = 0.7 V + (4 mA)(100 Ω) = 1.1 V

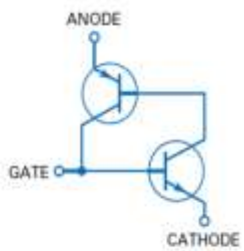
So, the input voltage has to be reduced from 15 V to slightly less than 1.1 V to open the diode.

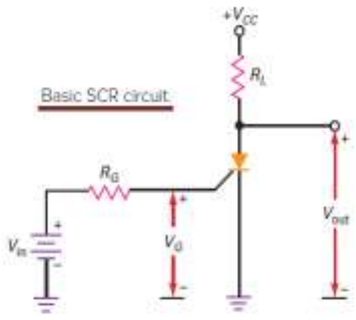
Silicon Controlled Rectifier (SCR)

The SCR is the most widely used Thyristor. It can switch very large currents ON and OFF. Because of this, it is used to control motors, ovens, air conditioners, and induction heaters.

Triggering the SCR (turning ON)

A new input terminal called gate is added to the base of Q2 and by applying a trigger (sharp pulse) to the base of Q2, the SCR can be turned ON. Therefore, the SCR is equivalent to a latch with a trigger input. Since the gate of an SCR is connected to the base of an internal transistor, it takes at least 0.7 V to trigger an SCR. The trigger momentarily increases the base current of Q2.





This starts the positive feedback, which drives both transistors into saturation and voltage across the device is ideally zero.

Required Input Voltage

An SCR has a gate voltage V_G. When this voltage is more than V_{GT}, the SCR will turn on and the output voltage will drop from $+V_{CC}$ to a low value. The input voltage needed to trigger an SCR

- **Vin $\geq V_{GT} + I_{GT}R_G$**

In this equation, V_{GT} and I_{GT} are the gate trigger voltage and current for the device. Unless the above equation is satisfied, the SCR cannot turn on.

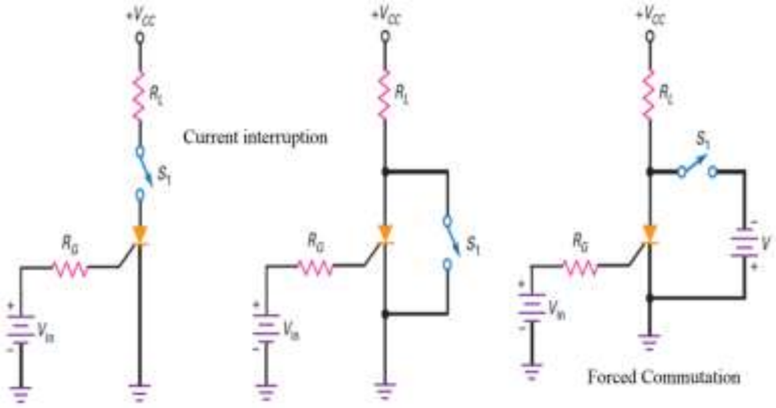
Resetting the SCR

After the SCR has turned on, it stays on even though the gate supply V_{in} is reduced to zero. In this case, the output remains low indefinitely.

To reset the SCR, reduce the anode to cathode current to a value less than its holding current I_H. This can be done by reducing V_{CC} to a low value.

Since the holding current flows through the load resistor. supply voltage for turn-off has to be less than:

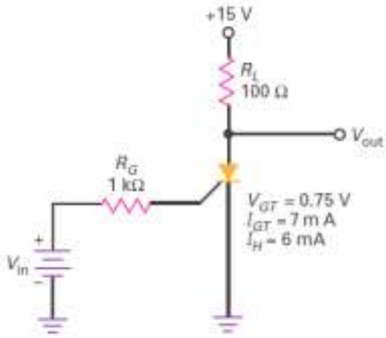
$\mathbf{V_{CC} = 0.7\ V + I_H R_L}$



The other two common methods are current interruption and forced commutation.

Current interruption- By either opening the series switch, or closing the parallel switch The anode-to-cathode current will drop down below its holding current value and the SCR will switch to its off state.

Forced commutation - When the switch is depressed, a negative V_{AK} voltage is momentarily applied. This reduces the forward anode-to-cathode current below I_H and turns off the SCR.



Problem - The SCR has a trigger voltage of 0.75 V and a trigger current of 7 mA. What is the input voltage that turns the SCR on? If the holding current is 6 mA, what is the supply voltage that turns it off?

- **SOLUTION**

The minimum input voltage needed to trigger the SCR is:

$V_{in} = V_{GT} + I_H R_L$

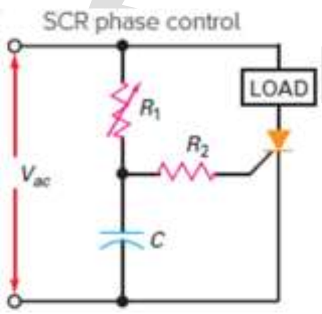
- $V_{in} = 0.75 + (7\ mA)(1\ k\Omega) = 7.75\ V$

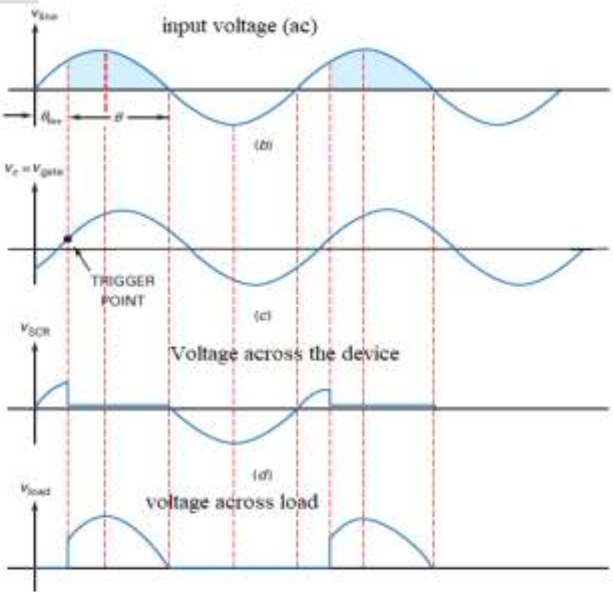
The supply voltage that turns off the SCR is:

- $V_{CC} = V_{GT} + I_{GT} R_L = 0.7 + (6\ mA)(100\Omega) = 1.3\ V$

SCR Phase Control

Phase Control of SCR means having control on the phase relationship between the start of current through the SCR and source voltage. These are used as rectifiers to generate controlled DC voltage from the ac input.





The ac line voltage being applied to an SCR circuit that controls the current through a heavy load.

In this circuit, variable resistor R1 and capacitor C shift the phase angle of the gate signal. When R1 is zero, the gate voltage is in phase with the line voltage, and the SCR acts like a half-wave rectifier. R2 limits the gate current to a safe level.

When R1 increases, the ac gate voltage lags the line voltage by an angle between 0° and 90°. Before the trigger point the SCR is off and the load current is zero. At the trigger point, the capacitor voltage is large enough to trigger the SCR. When this happens, almost all of the line voltage appears across the load and the load current becomes high. Ideally, the SCR remains latched until the line voltage reverses polarity.

Calculations

Capacitive reactance $X_C = \dfrac{1}{2\pi f C}$

Impedance $Z_T = \sqrt{R^2 + X_C{}^2}$ and its phase angle $\theta_z = -\tan^{-1}\left(\dfrac{X_C}{R}\right)$

Using input voltage as reference,

the current through the capacitor is given by $I_C = \dfrac{V_{in}}{Z_T}$

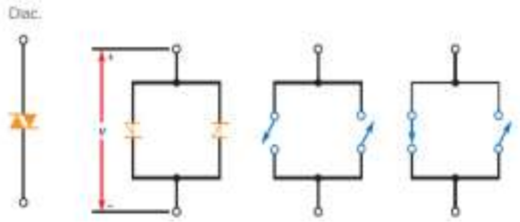
the voltage across the capacitor $V_C = (I_C \angle\theta)(X_C \angle -90)$

The amount of delayed phase shift will be the approximate firing angle of the circuit. The conduction angle is found by subtracting the firing angle from 180°.

Bidirectional Thyristors

The DIAC and Triac are bidirectional Thyristors. These devices can conduct in either direction.

DIAC - DIAC stands for Diode for Alternating Current.

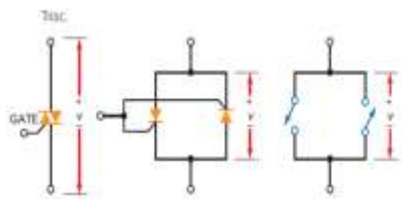


A DIAC is a semiconductor diode for conducting alternating current. Basically, it is a power electronic bidirectional semiconductor uncontrolled switch that is capable of conducting the electric current in both directions only after its break-over voltage, V_{BO}, has been reached momentarily.

The Diac is sometimes called a Silicon Bidirectional Switch (SBS)

TRIAC is basically a 3 terminal **ac switch** that shows **conduction in both the directions**. These are triggered into conduction by low energy gate signal.

TRIAC is a contraction of **TRI**ode for **A**lternating **C**urrent. It is a **bidirectional device** that belongs to the Thyristor family and is basically a Diac with gate terminal used to control the turn-on conditions of the device.



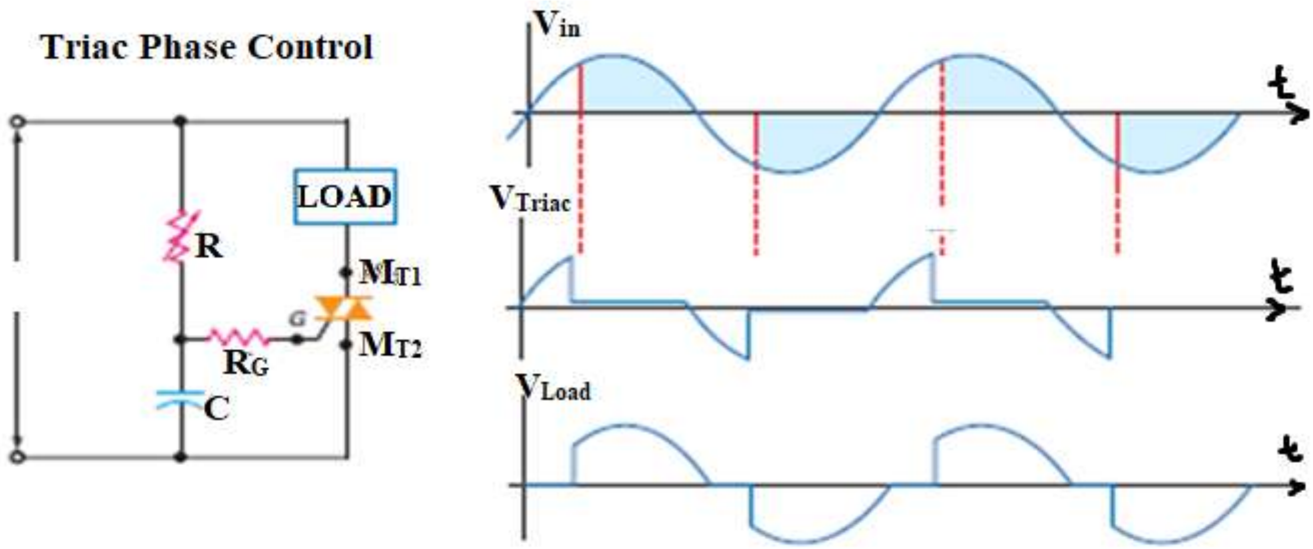
The Triac acts like two SCRs connected in reverse. Because of this, the Triac can control current in both directions. If v has the polarity shown a positive trigger will close the left latch. When **v** has opposite polarity, a negative trigger will close the right latch.

The Triac normally operates in quadrants I and III during typical ac applications.

Triac Phase Control

An RC circuit that varies the phase angle of the gate voltage to a triac. The circuit can control the current through a heavy load. When the capacitor voltage is large enough to supply the trigger current, the triac conducts. Once turned ON, the Triac continues to conduct until the line voltage returns to zero.

When conduction on both half-cycles is important, Triacs are useful devices, especially in industrial applications.

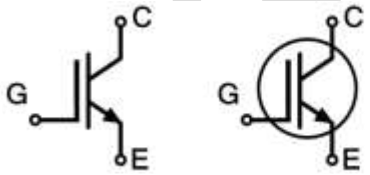


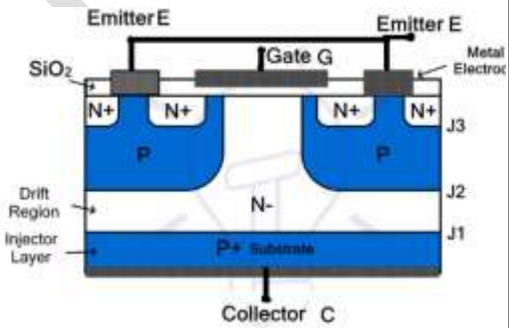
IGBT Insulated Gate Bipolar Transistor (IGBT)

The IGBT is the combination of BJT and MOSFET. It is a four-layer PNPN voltage controlled high speed switching device capable of handling large currents. It has three PN junctions and three terminals Gate (G), Collector(C) and Emitter (E). Gate terminal as it is the input part, taken from MOSFET while the collector and emitter as they are the output, taken from the BJT.

Construction

Figure shows schematic symbol of and internal structure IGBT.





IGBT is made of four layers of semiconductor to form a PNPN structure. The collector (C) electrode is attached to P layer while the emitter (E) is attached between the P and N layers. A P+ substrate is used for the construction of IGBT. An N- layer is placed on top of it to form PN junction J1. Two P regions are fabricated on top of N- layer to form PN junction J2. The P region is designed in such a way to leave a path in the middle for the gate (G) electrode. N+ regions are diffused over the P region as shown in the figure.

The emitter and gate are metal electrodes. The emitter is directly attached to the N+ region while the gate is insulated using a silicon dioxide layer. The base P+ layer inject holes into N- layer that is why it is called injector layer. While the N- layer is called the drift region

The N- layer is designed to have a path for current flow between the emitter and collector through the junction using the channel that is created under the influence of the voltage at the gate electrode.

Working – The collector-emitter is connected to V_{CC} such that the collector is kept at a positive voltage than the emitter. The junction j_1 becomes forward biased and j_2 becomes reverse biased. At this point, there is no voltage at the gate. Due to reverse j_2, the IGBT remains switched off and no current will flow between collector and emitter.

Applying a gate voltage V_G positive than the emitter, negative charges will accumulate right beneath the SiO_2 layer due to capacitance effect. Increasing the V_G increases the number of charges which eventually form a layer when the V_G exceeds the threshold voltage, in the upper P-region. This layer form N-channel that shorts N- drift region and N+ region.

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The electrons from the emitter flow from N+ region into N- drift region. While the holes from the collector are injected from the P+ region into the N- drift region. Due to the excess of both electrons and holes in the drift region, its conductivity increase and starts the conduction of current. Hence the IGBT switches ON.

Advantages

(i) It has higher voltage and current handling capability (ii) It has very high input impedance and can switch high currents using very low voltage. (iii) It is voltage-controlled device with very low ON-state resistance i.e. it has no input current and low input losses. (iv) The gate drive circuitry is simple and cheap.
(v) It can be easily switched ON by applying positive voltage and OFF by applying zero or slightly negative voltage. (vi) It has a higher power gain than both BJT and MOSFET.

Disadvantages

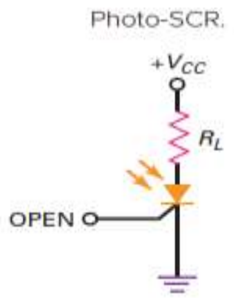
(i) It has a moderate switching speed. (ii) It is unidirectional it cannot conduct in reverse. (iii)It cannot block higher reverse voltage. (iv)It is costlier than BJT and MOSFET.

Applications of IGBT

IGBTs are used in AC as well as DC circuits. It is used in SMPS (Switched Mode Power Supply), UPS (Uninterruptible Power Supply) systems. It is used in AC and DC motor drives offering speed control. It is used in chopper and inverters.

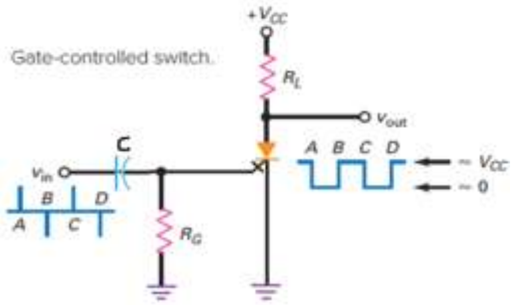
Other Thyristors

Photo-SCR



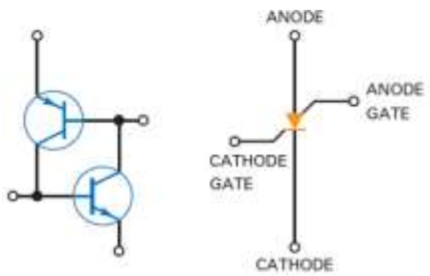
A photo-SCR, also known as a light-activated SCR. The arrows represent incoming light that passes through a window and hits the depletion layers.
When the light is strong enough, valence electrons are dislodge from their orbits and become free electrons. The flow of free electrons starts the positive feedback, and the photo-SCR closes. After a light trigger has closed the photo-SCR, it remains closed, even though the light disappears. For maximum sensitivity to light, the gate is left open. To get an adjustable trip point, trigger adjust is included. The resistance between the gate and ground diverts some of the light-produced electrons and reduces the sensitivity of the circuit to the incoming light.

Gate-Controlled Switch



Low-current drop-out is the normal way to open an SCR. But the gate-controlled switch is designed for easy opening with a reverse biased trigger. A gate-controlled switch is closed by a positive trigger and opened by a negative trigger. Each positive trigger closes the gate-controlled switch, and each negative trigger opens it. Because of this, he output is a square-wave .
The gate-controlled switch has been used in counters, digital circuits, and other applications in which a negative trigger is available.

Silicon Controlled Switch



It is latch with access to both bases. A forward-bias trigger on either base will close the silicon controlled switch. Likewise, a reverse-bias trigger on either base will open the device. The lower gate is called the *cathode gate*, and the upper gate is the *anode gate*. The silicon controlled switch is a low-power device compared to the SCR. It handles currents in milli-amperes rather than amperes.

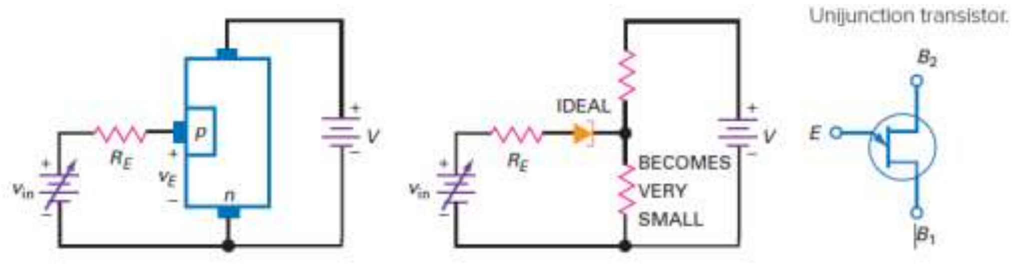
Uni-junction Transistor (UJT)

A UJT is a three terminal semiconductor device which exhibits negative resistance and switching characteristics for use as relaxation oscillator in phase control applications.

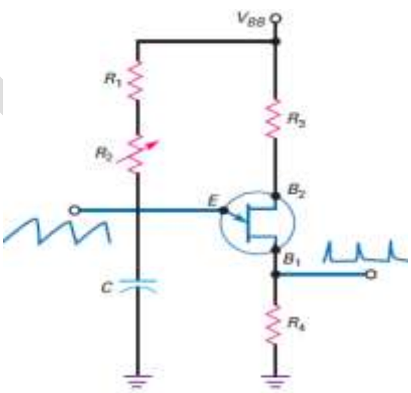
Construction of a UJT

The basic structure of a UJT, its symbol and equivalent circuit is shown in figure. It essentially consists of a lightly-doped N-type silicon bar with a small piece of heavily doped P-type material alloyed to its one side to produce single P-N junction. The silicon bar, at its ends, has two ohmic contacts designated as base-1 (B_1) and base-2 (B_2) and the P-type region is termed the emitter (E). The emitter junction is usually located closer to base-2 (B_2) than base-1 (B_1) so that the device is not symmetrical.

When the input voltage is zero, the device is non-conducting. When the input voltage is increased above the standoff voltage (given on a data sheet), the resistance between the p region and the lower n region becomes very small.



UJT relaxation Oscillator - The UJT can be used to form a pulse-generating circuit called a UJT relaxation oscillator. When the power supply is applied, the capacitor charges towards V_{BB} with a time constant $(R_1 + R_2)C$. When the capacitor voltage reaches a value equal to the standoff voltage, the UJT turns ON. The internal lower base (lower n region) resistance quickly drops in value allowing the capacitor to discharge. The capacitor discharge continues until low-current drop-out occurs. When this happens, the UJT turns OFF. Now the capacitor begins to charge again and the cycle repeats. The charging RC time constant is significantly larger than the discharge time constant. Hence a sharp pulse waveform developed across the external resistor at B_1 which can be used as a trigger source for controlling the conduction angle of SCR and Triac circuits. The waveform developed across the capacitor can be used in applications where a saw-tooth generator is needed.



The period of the waveform is given by $t = RC\ln\left(\frac{1}{(1-\eta)}\right)$ sec

where η is the intrinsic standoff ratio and usually $\eta = 0.5$ and $R = R_1 + R_2$

hence the frequency of the saw-tooth waveform is $f_o = \frac{1.5}{RC}$ Hz

Review Questions

1. What is a Thyristor? Mention few applications of Thyristors.
2. With proper illustrations, explain the operation of Four layered diode.
3. Explain the terms Break-over and Low Current drop out with respect to Four layered diode.
4. With a neat diagram explain the operation Silicon Controlled Rectifier (SCR).
5. What is Phase control? With a neat circuit diagram and equations, explain SCR phase control circuit
6. Define the terms (i) Holding current and (ii) Latching current
7. Write short notes on (i) Resetting SCR (Commutation) (ii) Diac (iii) Triac
8. With a neat diagram, explain the operation of IGBT
9. Write explanatory note on (i) Photo-SCR (ii) Gate Controlled Switch (iii) Silicon Controlled Switch
10. With a neat diagram, explain the working of UJT relaxation oscillator